

Six Degrees of Football: Network Science of Player Transfers in Europe's Top-5 Leagues

Mert Siraç Akkuş (34519) Taylan İrak (30702) Oğuz Temelli (32594)

Sami Efe Sarıbay (32044) Göktüğ Kırkıl (33864)

Sabancı University

Abstract—Professional football is one of the few global labor markets where career mobility is publicly documented over long historical periods. In this study, we model European elite-football mobility as a large-scale player co-tenancy network spanning the top five European leagues between 1992 and 2024. Using transfer records from more than 46,000 incoming transfers, we construct a weighted player graph in which nodes represent players and edges connect players who shared the same club during the same season. The resulting network contains 19,288 players and 350,819 weighted ties.

We analyze the graph using classical network-science methods including degree distributions, centrality measures, community detection, assortativity, k-core decomposition, temporal evolution, and link prediction. Results show that the transfer network exhibits strong small-world structure, combining very high clustering with short average path lengths. Nearly all players belong to a single giant connected component, and 99.998% of reachable player pairs are connected within six hops. Community detection reveals strong national clustering despite the globalization of football markets, while temporal analysis suggests that the 1995 Bosman ruling coincides with a sharp increase in network connectivity. Centrality analyses further indicate that highly mobile journeyman players, rather than globally famous superstars, form the structural backbone of the network.

Our findings demonstrate that European football constitutes a densely connected and highly navigable labor ecosystem whose topology reflects both institutional structure and historical regulatory change.

Index Terms—network science, football analytics, sports networks, small-world networks, player mobility, graph analysis, transfer networks

I. INTRODUCTION

Professional football has become a highly connected labor market. Every season, thousands of players move between clubs, leagues, and countries, creating a mobility network shaped by money, league rules, and transfer policies. Unlike many labor markets where mobility data are fragmented or inaccessible, football transfers are publicly documented in substantial detail over long time horizons. This makes football a useful case for network-science analysis.

Most football analytics work focuses on tactics, passing networks, player value, or match results. Comparatively fewer studies examine the structural topology of football labor mobility at continental scale. Understanding this topology is important because transfer patterns influence talent circulation, club interconnectedness, player exposure, and long-term market integration.

In this work, we model elite European football as a player co-tenancy network. Rather than directly connecting clubs through transfers, we connect players who shared the same club during the same season. This modeling choice captures professional overlap between players and allows the resulting graph to represent shared institutional environments. Edge weights correspond to repeated co-membership across multiple seasons.

Using transfer data from Europe's top five leagues between 1992 and 2024, we investigate the following research questions:

- Does elite European football exhibit small-world structure?
- How concentrated or modular is the transfer ecosystem?
- How did the Bosman ruling period affect network connectivity over time?
- Which players occupy structurally central positions?
- Can graph topology help predict future player co-tenancy relationships?

This paper makes three main contributions. First, we construct a large-scale weighted co-tenancy network representing more than three decades of elite European football. Second, we show that the resulting topology exhibits strong small-world characteristics while maintaining significant national modularity. Third, we demonstrate that major historical regulatory change, particularly the 1995 Bosman ruling, is temporally associated with rapid network integration.

II. RELATED WORK

Network science has been widely applied to social systems, biological systems, transportation systems, and online interaction networks. Classical studies by Watts and Strogatz demonstrated that many real-world systems simultaneously exhibit high clustering and short path lengths, producing small-world behavior [1]. Later work on scale-free networks showed that many complex systems display broad degree distributions and hub-like structures [2]. Newman provides a broad overview of methods used to analyze empirical networks, including centrality, clustering, components, and community structure [4].

Sports analytics has increasingly adopted graph-theoretic methods, especially in football. Many existing studies focus on passing networks during matches, tactical positioning,

possession structures, and team coordination. In these studies, players are usually represented as nodes and passes as directed weighted edges. Beyond match-level analysis, transfer networks and player mobility systems can also be studied using graph models. Club-level transfer graphs reveal market flows, while player-level co-membership graphs capture shared professional environments.

This study differs from many existing football-network analyses by constructing a player-level co-tenancy network rather than a match-level passing graph or club-to-club transfer graph. This allows us to examine the long-term topology of European football labor mobility. The work also connects network evolution to institutional change, particularly the Bosman ruling, which altered player mobility within European football.

III. DATASET AND PREPROCESSING

A. Data Source

The dataset used in this study was obtained from a publicly available football transfer dataset distributed under the MIT license. The dataset contains football transfer records from Europe's top five leagues: Premier League, La Liga, Bundesliga, Serie A, and Ligue 1. The observation window spans the 1992–2024 seasons.

After preprocessing and deduplication, the dataset contains 46,674 incoming transfer records. For each transfer entry, the dataset includes player name, club, season, age, position, transfer fee, transfer movement, league, and country.

B. Preprocessing

Transfer records often appear twice in football datasets: once as an outgoing transfer from the source club and once as an incoming transfer to the destination club. To avoid double-counting, we retain only records labeled as incoming transfers. Duplicate rows and incomplete entries are removed during preprocessing. Seasons are normalized into a unified format, and categorical attributes are standardized across leagues.

A limitation is that player identities are represented using names rather than unique persistent identifiers. Consequently, players with identical names may occasionally merge into a single node. This issue particularly affects common single-name football identifiers such as “Fernando” or “Felipe.” While the overall network structure remains informative, some centrality rankings may be affected by name collisions.

IV. GRAPH CONSTRUCTION AND METHODOLOGY

A. Bipartite Construction

We first construct a bipartite graph connecting players to club-season entities. Each club-season pair is represented as a unique node:

$$CS = (\text{club}, \text{season}). \quad (1)$$

An edge exists between a player node and a club-season node if the player belonged to that club during that season. The bipartite graph is then projected onto the player axis. Two

players are connected if they shared at least one club-season membership.

B. Weighted Projection

The projected graph is weighted. The edge weight between players i and j corresponds to the number of shared club-season memberships:

$$w_{ij} = |CS_i \cap CS_j|. \quad (2)$$

This weighting scheme captures repeated professional overlap and stronger institutional association. The resulting network contains 19,288 player nodes and 350,819 weighted edges.

C. Network Measures

We compute degree distribution, connected components, clustering coefficient, transitivity, assortativity, shortest-path statistics, Louvain communities, centrality measures, and k -core decomposition. Centrality measures include degree, betweenness, closeness, eigenvector centrality, and PageRank. Exact betweenness centrality is computationally expensive on a graph of this size, so we use approximation sampling with $k = 500$.

Power-law fitting follows the Clauset–Shalizi–Newman methodology [3]. We compare the pure power-law model against exponential, lognormal, and truncated alternatives.

Community detection is performed using the Louvain modularity-maximization algorithm [5]. Small-world structure is evaluated by comparing clustering coefficient and average shortest-path length against random and lattice reference graphs. For the random baseline, we generate Erdős–Rényi graphs while keeping the number of nodes and overall network density fixed. This allows a direct comparison between the real football network and a random graph with similar size characteristics. The lattice reference graph preserves strong local clustering but produces much longer path lengths. The small-world coefficient is defined as:

$$\sigma = \frac{C/C_{rand}}{L/L_{rand}}, \quad (3)$$

where C is clustering coefficient, L is average path length, and C_{rand} and L_{rand} correspond to random-graph baselines.

D. Temporal Analysis and Link Prediction

Temporal snapshots are generated season by season to analyze evolving network connectivity over time. We examine giant connected component share, clustering evolution, and average degree evolution, with special attention to the period surrounding the 1995 Bosman ruling.

We also evaluate whether local topology can predict future co-tenancy relationships. Training graphs are constructed using historical data before 2018, while future co-tenancy edges between 2019 and 2022 are used for evaluation. Features include Common Neighbours, Jaccard Coefficient, Adamic–Adar Index, and Preferential Attachment. These features are combined using logistic regression.

V. RESULTS

A. Global Network Structure

The football co-tenancy graph forms a highly connected system. The network contains 19,288 players, 350,819 weighted edges, and 2,429,012 triangles. The giant connected component contains approximately 99.89% of all players. Only a handful of very small disconnected components remain outside the main graph. The approximate network diameter is 7, while the estimated radius is 5. These findings indicate that elite European football behaves as a single large-scale interconnected labor ecosystem.

TABLE I
GLOBAL NETWORK STATISTICS.

Metric	Value
Players	19,288
Edges	350,819
Giant component share	99.89%
Triangles	2,429,012
Average clustering	0.684
Transitivity	0.324
Approx. diameter	7
Approx. radius	5
Assortativity	0.191
Max k-core	114

B. Degree Distribution

The degree distribution is strongly heavy-tailed. Most players exhibit relatively small numbers of co-tenancy relationships, while a smaller set of highly mobile players accumulate very large neighborhoods. Power-law fitting yields $\alpha \approx 2.83$, with more than 6,000 nodes located in the tail region. However, statistical comparisons indicate that lognormal and truncated alternatives outperform a pure power-law model. Therefore, the network should be interpreted as broad-scale and heavy-tailed rather than strictly scale-free.

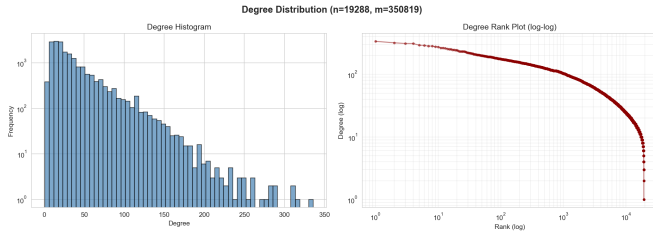


Fig. 1. Degree distribution of the player co-tenancy network. The distribution is heavy-tailed, but alternative distributions fit better than a pure power-law model.

C. Small-World Structure

The network exhibits strong small-world behavior. The average local clustering coefficient is approximately $C = 0.684$, while the corresponding random-graph clustering coefficient is approximately $C_{rand} = 0.00185$. This means that the real football network is nearly 370 times more clustered than a random network with the same number of nodes and density. At

the same time, the average shortest-path length remains small at approximately $L = 3.49$, close to the random baseline. The resulting small-world coefficient is approximately $\sigma = 319$, indicating strong small-world organization.

The football network maintains clustering values close to the lattice reference while preserving path lengths close to random graphs. Structurally, this happens because club rosters create dense local groups, while transfers create shortcuts between distant parts of the graph.

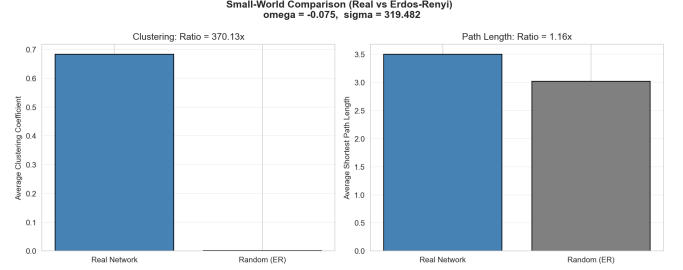


Fig. 2. Small-world comparison between the real network, random baseline, and lattice-like reference graph. The real graph combines high clustering with short path length.

D. Six Degrees of Football

The network also supports the six-degrees idea. Approximately 99.998% of reachable player pairs are connected within six hops or fewer. The maximum observed shortest-path distance is seven hops. This shows that elite football careers are much more connected than they may seem.

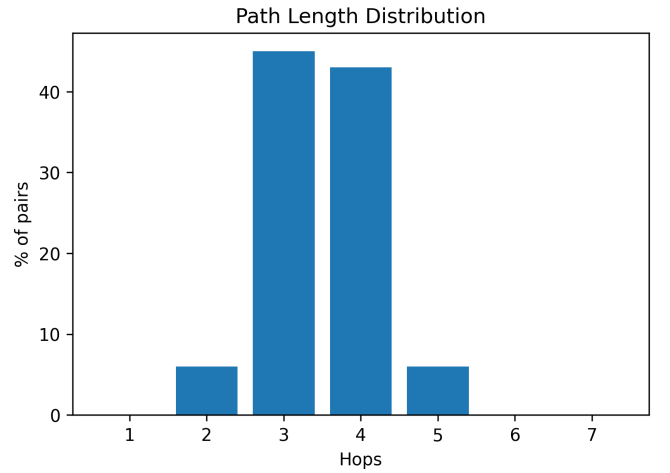


Fig. 3. Shortest-path length distribution. Nearly all reachable player pairs are connected within six hops.

E. Community Structure

Louvain community detection identifies 21 major communities with modularity $Q \approx 0.61$, indicating strong modular organization. The largest communities align heavily with national structures. The five largest communities correspond predominantly to Italy, Germany, France, England, and Spain,

each with more than 94% national consistency. Despite increasing globalization, domestic league structures therefore remain strongly visible in the topology.

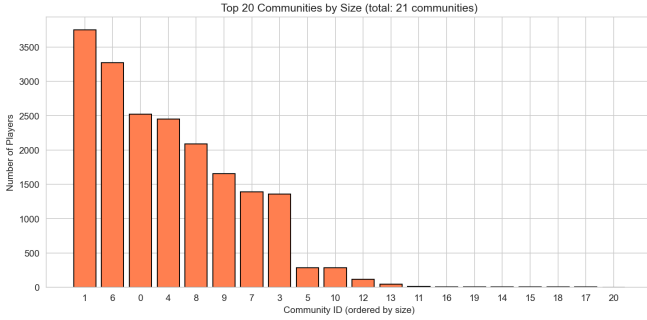


Fig. 4. Largest Louvain communities. The biggest communities align strongly with national league structures.

F. Centrality Analysis

Centrality rankings reveal that highly mobile journeyman players dominate structural importance measures. Players such as Fernando, Antonio Cassano, Gaël Kakuta, and Marco Borriello consistently rank highly across betweenness, PageRank, degree, and eigenvector centrality. This result suggests that structural importance in football mobility networks is driven more by transfer diversity and institutional reach than by global celebrity status.

TABLE II
REPRESENTATIVE TOP CENTRAL PLAYERS BY METRIC.

Metric	Leading Players
Degree	Felipe; Antonio Cassano; Gaël Kakuta
Betweenness	Fernando; Júlio César; Gaël Kakuta
PageRank	Fernando; Gaël Kakuta; Marco Borriello
Eigenvector	Antonio Cassano; Mattia Cassani; Ezequiel Schelotto

G. K-Core Structure

K-core decomposition reveals a dense inner nucleus. The maximum k-core value is $k_{max} = 114$, and the innermost core contains 115 players. Every player in this inner group shares at least 114 co-tenancy relationships with the rest of the core. This structure represents an elite mobility backbone formed primarily by extensively traveled professional players. In other words, the deepest core is mainly formed by highly mobile multi-league players rather than only globally dominant stars.

H. Temporal Evolution and the Bosman Ruling

Temporal analysis reveals rapid network integration during the mid-1990s. The giant connected component expands dramatically from approximately 6% of players in early seasons to nearly complete connectivity after 1995. This transition coincides temporally with the Bosman ruling, which significantly liberalized player mobility within the European Union. While the analysis does not establish direct causality, the timing

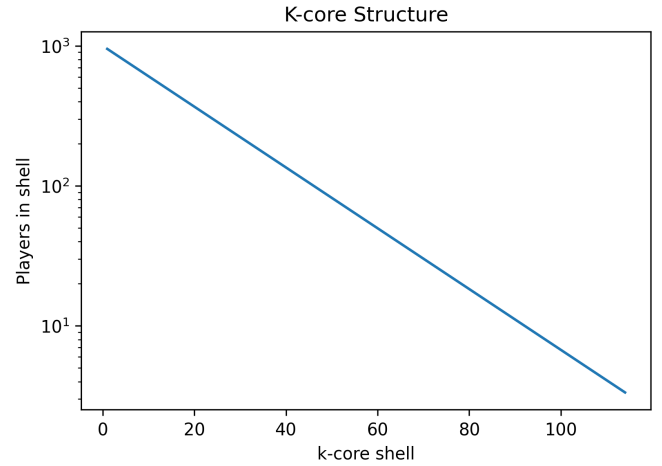


Fig. 5. K-core shell-size distribution. The densest inner core contains 115 players at $k = 114$.

suggests that institutional policy changes may have contributed to increased transfer-network integration.

At the same time, average degree increases steadily, clustering decreases gradually, and density declines as the network expands. These trends are consistent with large-scale network growth and increased international labor circulation.

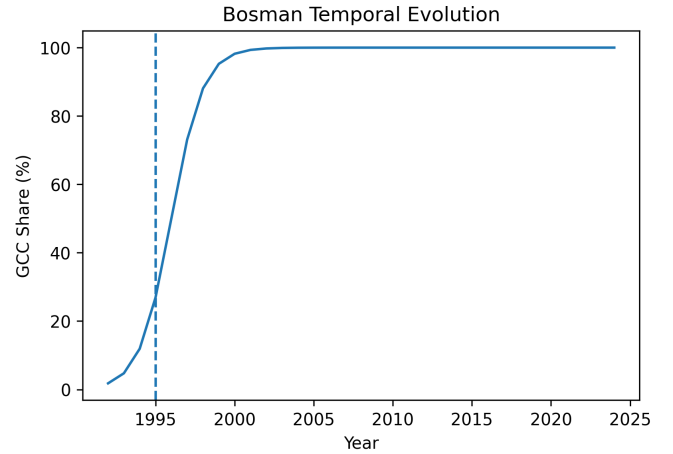


Fig. 6. Temporal evolution of the giant connected component. The sharp increase in connectivity aligns with the period around the 1995 Bosman ruling.

I. Link Prediction

Link-prediction experiments show that local graph topology contains predictive information regarding future co-tenancy relationships. Classical neighborhood-based heuristics combined with logistic regression provide useful signals for identifying future player overlap. In particular, Common Neighbours, Adamic–Adar, and Preferential Attachment capture structurally meaningful patterns. These results suggest that football labor mobility exhibits partially predictable structural regularities.

VI. DISCUSSION

Overall, the results show that elite European football works like one connected labor network while still keeping strong local and national structure. The small-world pattern makes sense because club rosters create tight local groups, while transfers create links between distant parts of the graph. Each club season effectively creates a dense local clique, while transfers create shortcuts connecting distant regions of the network.

The strong modularity observed in community detection demonstrates that national structure remains highly influential despite increasing globalization. Domestic leagues continue to function as partially separated mobility submarkets. At the same time, very short average path lengths reveal substantial integration between these submarkets.

The centrality results are also important. Structural importance is not dominated by globally famous superstars but rather by highly mobile players who repeatedly bridge clubs, leagues, and communities. These players act as mobility connectors that compress network distances and facilitate large-scale connectivity.

The Bosman analysis further illustrates how institutional policy can reshape labor-market topology. The rapid expansion of the giant connected component after the 1995 ruling suggests that regulatory liberalization may have accelerated cross-club integration across European football. Temporal analysis also indicates that the 2010s produced the largest growth in new edges, suggesting that player mobility increased further during the modern transfer-market era.

VII. LIMITATIONS

Several limitations should be acknowledged. First, the dataset represents only Europe's top five leagues and therefore excludes lower divisions, youth systems, and many international football markets. Consequently, the observed topology primarily reflects elite European football rather than global football as a whole.

Second, player identity is represented using names rather than persistent identifiers. Players with identical names may therefore merge into single nodes, potentially inflating some centrality measures.

Third, co-tenancy does not necessarily imply direct interaction. Players may share a club-season while experiencing different playing times, injuries, or loan arrangements. Fourth, transfer records themselves may contain historical incompleteness or reporting inconsistencies, particularly for earlier decades. Finally, the link-prediction experiments remain relatively simple and primarily demonstrate proof-of-concept predictive capability rather than optimized forecasting performance.

VIII. CONCLUSION

In this study, we constructed and analyzed a large-scale football co-tenancy network spanning Europe's top five leagues between 1992 and 2024. The resulting graph exhibits strong small-world structure characterized by high clustering, short

path lengths, dense local communities, and near-complete global connectivity.

We further showed that elite football forms a highly integrated labor ecosystem, national structures remain visible through strong modularity, highly mobile journeyman players act as structural bridges, and temporal connectivity growth aligns with the Bosman ruling period. Overall, football mobility networks are a useful setting for network science because they show how rules, clubs, and player movement shape a large professional system. The main small-world mechanism is simple: each club-season creates a dense local group, and transfers connect these groups across leagues and years.

REFERENCES

- [1] D. J. Watts and S. H. Strogatz, "Collective dynamics of 'small-world' networks," *Nature*, vol. 393, no. 6684, pp. 440–442, 1998.
- [2] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, 1999.
- [3] A. Clauset, C. R. Shalizi, and M. E. J. Newman, "Power-law distributions in empirical data," *SIAM Review*, vol. 51, no. 4, pp. 661–703, 2009.
- [4] M. E. J. Newman, *Networks: An Introduction*. Oxford, U.K.: Oxford University Press, 2010.
- [5] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2008, no. 10, 2008.
- [6] S. Boccaletti, V. Latora, Y. Moreno, M. Chavez, and D.-U. Hwang, "Complex networks: Structure and dynamics," *Physics Reports*, vol. 424, no. 4–5, pp. 175–308, 2006.
- [7] L. C. Freeman, "Centrality in social networks conceptual clarification," *Social Networks*, vol. 1, no. 3, pp. 215–239, 1978.
- [8] P. Erdős and A. Rényi, "On random graphs," *Publicationes Mathematicae*, vol. 6, pp. 290–297, 1959.
- [9] S. Brin and L. Page, "The anatomy of a large-scale hypertextual Web search engine," *Computer Networks and ISDN Systems*, vol. 30, no. 1–7, pp. 107–117, 1998.
- [10] Court of Justice of the European Union, "Union Royale Belge des Sociétés de Football Association ASBL v Jean-Marc Bosman," Case C-415/93, 1995.